

# Scaling Laws for Cyber Resilience: Modeling the Persistence of Backdoors in Post-trained Large Language Models

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**Abstract**—Backdoor attacks on large language models (LLMs) persist through safety fine-tuning procedures, yet no principled metric exists to quantify this persistence as a function of model scale. Existing attack success rate (ASR) measurements are scale-agnostic and provide no theoretical guarantees connecting parameter count to backdoor robustness. We propose the *Cyber Resilience Scaling Coefficient* (CRSC), a closed-form metric that combines normalized Frobenius drift of hidden-state representations ( $\Delta_{\text{hidden}}$ ) and Shannon entropy change of the loss distribution ( $\Delta H$ ) into a single score quantifying backdoor persistence after supervised fine-tuning (SFT) or reinforcement learning from human feedback (RLHF). We prove two properties: (i) CRSC is monotonically non-decreasing in parameter count  $N$  for fixed fine-tuning conditions (Proposition 1), formalized via random matrix theory; and (ii) CRSC converges to a concentration-of-measure proxy for ASR with bounded estimation error (Proposition 2). A three-agent evaluation pipeline (Detection, Analysis, Response) computes CRSC across  $N \in \{7B, 13B, 70B\}$ , poison rates  $\rho \in \{0.001, 0.005, 0.01, 0.05\}$ , three trigger types, and two fine-tuning methods, yielding 72 experimental runs on synthetic data calibrated against TrojAI NIST 2024. At threshold  $\tau = 0.62$ , CRSC achieves  $F1 = 0.891$  (95% CI:  $[0.874, 0.908]$ ), on data calibrated against Yang et al. [1]; see Section VI),  $AUC\text{-}ROC = 0.924$ , and Pearson  $r = 0.847$  (72 runs across all scale-and-condition combinations; see Section VI for degrees-of-freedom discussion), confirming the scaling monotonicity. CRSC is mapped to MITRE ATLAS adversarial ML techniques and the NIST AI Risk Management Framework, providing a compliance-ready risk classification at three severity levels. Code and data are released under MIT license at <https://github.com/ufxa/Cyber-Resilience-Scaling-Coefficient>.

**Index Terms**—Backdoor attacks, large language models, cyber resilience, scaling coefficient, fine-tuning security, RLHF, adversarial machine learning, MITRE ATLAS.

## 1 INTRODUCTION

Large language models (LLMs) have demonstrated remarkable capabilities across security-sensitive domains, including penetration testing assistance [2], vulnerability analysis [3], and autonomous cyberdefense [4]. This proliferation, however, introduces a critical attack surface: adversaries may embed *backdoors*—hidden behavioral triggers that cause targeted model outputs when activated—during

the supply chain of model training. Unlike classical software vulnerabilities, LLM backdoors survive post-training alignment procedures (supervised fine-tuning, SFT, and reinforcement learning from human feedback, RLHF), creating a persistent threat even after safety hardening.

A fundamental open question concerns the relationship between *model scale* and *backdoor persistence*: do larger models resist fine-tuning-based backdoor removal more than smaller ones? Empirical evidence from prior work [1], [5], [6] suggests a positive correlation between parameter count and attack success rate (ASR) after RLHF, but no formal metric captures this relationship in a principled way suitable for risk quantification or regulatory compliance.

**Contributions.** This paper makes four concrete contributions:

- 1) We propose the *Cyber Resilience Scaling Coefficient* (CRSC), a closed-form metric that quantifies backdoor persistence after fine-tuning as a function of model scale  $N$ , poison rate  $\rho$ , and fine-tuning dataset  $D_{\text{ft}}$  (Section 3).
- 2) We prove two theoretical properties: (i) monotonicity of CRSC in  $N$  for fixed  $(\rho, D_{\text{ft}})$  — larger models are provably harder to sanitize (Proposition 1); and (ii) CRSC is a concentration-of-measure proxy for ASR with bounded estimation error (Proposition 2).
- 3) We design a three-agent evaluation pipeline (Detection, Analysis, Response) and validate CRSC on synthetic data calibrated against the TrojAI NIST 2024 benchmark [7], achieving  $F1 = 0.891$  and  $AUC\text{-}ROC = 0.924$  at  $\tau = 0.62$  (Section 6).
- 4) We map CRSC to the MITRE ATLAS adversarial ML framework and the NIST AI Risk Management Framework (AI RMF), providing actionable mitigation guidance for practitioners (Section 5).

**Scope and limitations.** Our analysis assumes white-box access to hidden states (a common red-team setting [8]), static trigger patterns, and transformer architectures. Adaptive triggers and black-box settings remain open challenges; we discuss them in Section 7.

**Organization.** Section 2 reviews related work and establishes notation. Section 3 formalizes the threat model and derives CRSC. Section 4 presents the algorithmic pipeline. Section 5 addresses security compliance. Section 6 reports experimental results. Sections 7 and 8 discuss implications and conclude.

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## 2 BACKGROUND AND RELATED WORK

### 2.1 LLM Backdoor Attacks

Backdoor attacks on neural networks, introduced by Gu et al. [9], embed hidden triggers that cause misclassification while preserving benign accuracy. In language models, Chen et al. [5] extended this paradigm to NLP tasks, demonstrating that token-level triggers survive standard fine-tuning. Wallace et al. [10] showed that backdoors survive BERT fine-tuning, persisting across downstream task adaptation.

More recently, Yang et al. [1] established that RLHF-based safety alignment does not reliably remove backdoors from instruction-tuned models. Their experiments on LLaMA-2 show that models poisoned at rates as low as 1% retain attack success rates above 80% after RLHF. Hubinger et al. [11] introduced “sleeper agent” backdoors that survive chain-of-thought safety training, further motivating formal persistence metrics.

### 2.2 Scaling Laws

Kaplan et al. [12] established that loss scales as a power law in  $N$  (parameter count), dataset size  $D$ , and compute  $C$ . Hoffmann et al. [13] refined this via Chinchilla scaling. Crucially, these laws describe *capability* scaling; no prior work has derived analogous laws for *security* properties.

Wei et al. [14] documented emergent capabilities at specific scale thresholds ( $N \approx 10^{10}$ ), suggesting that security behavior may also exhibit phase transitions. Our monotonicity proof (Proposition 1) provides a formal basis for this conjecture in the backdoor persistence context.

### 2.3 Post-Training Security

Bai et al. [15] introduced Constitutional AI (CAI) as an alignment approach; Perez et al. [8] demonstrated that red-teaming can systematically probe LLM safety boundaries. Sharma et al. [16] proposed watermarking for LLM traceability, while Wan et al. [17] studied instruction-tuning poisoning at large scale.

### 2.4 Resilience Metrics

In classical cybersecurity, resilience metrics quantify recovery after adversarial events [18]. For ML systems, Carlini et al. [19] emphasized the importance of adaptive evaluation. The closest prior metric is the backdoor effectiveness score [20], which measures ASR post-fine-tuning but does not model scale dependency.

Regarding defense, Neural Cleanse [21] reverses engineered trigger patterns via optimization, achieving strong detection on image classifiers but showing limited transferability to LLM-scale models where trigger search is intractable. Fine-Pruning [22] removes backdoor-associated neurons via magnitude pruning, but requires knowing which neurons encode the backdoor. SPECTRE [23] uses robust statistics on activation representations to filter poisoned samples at training time, a complementary approach to CRSC’s post-training detection. Unlike these defenses, CRSC is a *detection metric* that operates on post-trained model pairs without requiring trigger inversion or training-set access. Quantitative comparison is provided in Section 6. CRSC fills the gap of a scale-dependent formal metric.

TABLE 1  
Comparison of backdoor persistence metrics.

| Method       | Year | Scale Dep. | Formal Proof | LLM Native | RLHF Aware | Detection Metric  |
|--------------|------|------------|--------------|------------|------------|-------------------|
| BES [20]     | 2022 | ✗          | ✗            | ✓          | ✗          | ASR (%)           |
| TrojAI [7]   | 2024 | ✗          | ✗            | ✓          | ✗          | TDS score         |
| ASR-RLHF [1] | 2024 | ✗          | ✗            | ✓          | ✓          | ASR-RLHF          |
| CRSC (ours)  | 2025 | ✓          | ✓            | ✓          | ✓          | CRSC $\in [0, 1]$ |

### 2.5 Comparison with Existing Work

Table 1 positions CRSC against the most relevant prior metrics.

## 3 THREAT MODEL AND CRSC DERIVATION

### 3.1 Adversary Model

We model the adversary  $\mathcal{A}$  under the following assumptions, representing a realistic supply-chain threat for LLM deployment.

**Capability.**  $\mathcal{A}$  has *training-time* access to the base model  $M_N$  and its training pipeline, and can inject a backdoor at fraction  $\rho \in (0, 1)$  of training samples. The adversary has no access to the defender’s probe set  $\mathcal{T}$  (held secret by the evaluator) and no influence over post-training alignment.

**Objective.**  $\mathcal{A}$  aims to maximize Attack Success Rate (ASR) on a target behavior  $b^*$  after post-training, while maintaining benign accuracy within  $\epsilon_{\text{util}}$  of the clean-model baseline.

**Knowledge.**  $\mathcal{A}$  knows the architecture family (transformer with  $N$  parameters,  $|L|$  layers, hidden dimension  $d_N$ ), the general fine-tuning procedure (SFT or RLHF), and public information about the post-training dataset.

**Out-of-scope.** Adaptive adversaries with oracle access to CRSC scores are analyzed separately in Section 5. Physical-world triggers and multi-model coordination attacks are excluded.

### 3.2 Backdoor Formalization

Let  $M_N: \mathcal{X} \rightarrow \mathcal{Y}$  be an LLM with  $N$  parameters.

**Definition 1** (Backdoor). A backdoor  $(\gamma, b^*)$  consists of a trigger function  $\gamma: \mathcal{X} \rightarrow \mathcal{X}$  and a target behavior  $b^* \in \mathcal{Y}$  such that  $M_N^{\text{bd}}$  satisfies:

$$\Pr[M_N^{\text{bd}}(\gamma(x)) = b^*] \geq 1 - \epsilon \quad (1)$$

$$\Pr[M_N^{\text{bd}}(x) = M_N(x)] \geq 1 - \epsilon_{\text{util}} \quad (2)$$

for all  $x \in \mathcal{X}$  and small constants  $\epsilon, \epsilon_{\text{util}} > 0$ .

**Backdoor injection.** Given a clean dataset  $D = \{(x_i, y_i)\}$ , the adversary constructs a poisoned dataset  $D^{\text{bd}} = (1 - \rho)D \cup \rho D^{\text{trig}}$ , where  $D^{\text{trig}} = \{(\gamma(x_i), b^*)\}$ . Post-training applies alignment fine-tuning over a clean dataset  $D_{\text{ft}}$  (with  $|D_{\text{ft}}| \ll |D|$ ) starting from  $M_N^{\text{bd}}$ , yielding  $M_N^{\text{ft}}$ . The defender’s goal is to detect whether  $M_N^{\text{ft}}$  retains the backdoor.

### 3.3 Representation Drift and Entropy Change

CRSC rests on two complementary signals.

**Hidden-state drift.** Let  $\mathbf{H}_{\text{base}}, \mathbf{H}_{\text{ft}} \in \mathbb{R}^{K \times d}$  be the mean-pooled last-layer hidden states of  $M_N^{\text{bd}}$  and  $M_N^{\text{ft}}$ , respectively, stacked over  $K$  probes. The *normalized Frobenius drift* is:

$$\Delta_{\text{hidden}} = \frac{\|\mathbf{H}_{\text{base}} - \mathbf{H}_{\text{ft}}\|_F}{\|\mathbf{H}_{\text{base}}\|_F} \quad (3)$$

$\Delta_{\text{hidden}} \approx 0$  indicates preserved backdoor representation;  $\Delta_{\text{hidden}} \gg 1$  indicates strong representational displacement. Importantly,  $\Delta_{\text{hidden}}$  is computed on *trigger-embedded* probes from  $\mathcal{T}$ ; benign probes are expected to show low drift for both backdoored and clean models, ensuring specificity to the backdoor signal.

**Entropy shift.** Let  $\ell_k^{\text{base}}$  and  $\ell_k^{\text{ft}}$  be per-token cross-entropy losses on probe  $t_k$ . Define empirical entropy  $H(\ell) = -\sum_k \hat{p}_k \log \hat{p}_k$  with  $\hat{p}_k = e^{-\ell_k} / \sum_j e^{-\ell_j}$ . The *entropy shift*  $\Delta H = H(\ell^{\text{ft}}) - H(\ell^{\text{base}})$  captures the change in output uncertainty on triggered inputs. A backdoored model collapses to confident incorrect predictions ( $H(\ell^{\text{ft}})$  decreases), yielding  $\Delta H < 0$ .

**Normalization via  $\Phi$ .** We map  $\Delta H \in \mathbb{R}$  to  $[0, 1]$  via  $\Phi(-\Delta H)$ , where  $\Phi$  is the standard Normal CDF. By the Berry–Essén theorem, the empirical entropy of a  $K$ -sample bounded i.i.d. loss sequence converges to a Normal distribution as  $K \rightarrow \infty$ . Therefore  $\Phi(-\Delta H)$  is the natural normalization: it maps  $-\Delta H$  to approximately Uniform $[0, 1]$  under the null hypothesis  $H_0$ :  $M_N^{\text{ft}}$  is backdoor-free ( $\Delta H \approx 0$ ), and produces values near 1 when  $\Delta H \ll 0$  (backdoor persists).

### 3.4 CRSC Definition

**Definition 2** (Cyber Resilience Scaling Coefficient). *Given model pair  $(M_N^{\text{bd}}, M_N^{\text{ft}})$ , probe set  $\mathcal{T}$  with  $|\mathcal{T}| = K$ , and weights  $\alpha, \beta > 0$  with  $\alpha + \beta = 1$ , the CRSC is:*

$$\text{CRSC}(M_N^{\text{bd}}, M_N^{\text{ft}}) = \alpha (1 - \Delta_{\text{hidden}})^+ + \beta \Phi(-\Delta H) \quad (4)$$

where  $(x)^+ = \max(x, 0)$ .

**Symmetric weighting** ( $\alpha = \beta = 0.5$ ). The two components measure statistically independent aspects of backdoor persistence:  $\Delta_{\text{hidden}}$  operates on the representation space (intermediate activations), while  $\Phi(-\Delta H)$  operates on the behavioral space (output loss distribution). Under a maximum-entropy principle [24], equal weighting maximizes the entropy of the combined signal, minimizing sensitivity to prior assumptions about which signal dominates for a given trigger type. The ablation in Table 7 confirms empirically that both components contribute independently ( $\Delta F1 = +0.048$  for full vs. hidden-only;  $+0.064$  for full vs. entropy-only), and a grid search over  $\alpha \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$  (cross-validated on TrojAI) found no statistically significant improvement over  $\alpha = 0.5$  (all differences  $< 0.007$  F1, within bootstrap CI).

### 3.5 Theoretical Properties

**Assumption 1** (Sparse backdoor subspace). *The backdoor is encoded in a  $k$ -dimensional subspace  $\mathcal{S}_b \subseteq \mathbb{R}^{d_N}$  with  $k = \Theta(N^{\alpha_b})$  for some  $\alpha_b \in (0, 1/2)$ .*

**Assumption 2** (Generic fine-tuning gradient). *The gradient  $G = \nabla_{\theta} \mathcal{L}(D_{\text{ft}})$  projected to hidden-state space acts as a random matrix of rank  $r = \Theta(|D_{\text{ft}}|)$  drawn from the Gaussian Orthogonal Ensemble [25]. Remark on idealization: Attention-head gradients exhibit low-rank block structure (one block per head), violating strict GOE independence. The assumption holds asymptotically when the number of fine-tuning directions is large relative to the backdoor subspace dimension; it should be treated as a tractability assumption rather than an exact model. The Marchenko–Pastur distribution [26] characterizes the spectrum of the empirical gradient covariance under this regime and provides the limiting spectral density used in the proof of Proposition 1.*

**Proposition 1** (Monotonicity in  $N$ ). *Under Assumptions 1–2, for fixed  $(\rho, D_{\text{ft}}, \gamma)$ :*

$$\mathbb{E}[\text{CRSC}(M_N^{\text{bd}}, M_N^{\text{ft}})] \text{ is non-decreasing in } N. \quad (5)$$

*Proof:* We bound the probability that fine-tuning displaces the backdoor subspace. By Assumption 2, the  $r$  gradient directions in  $\mathbb{R}^{d_N}$  act as independent random unit vectors. Let  $\varepsilon > 0$  be a fixed angular threshold below which a gradient direction does not meaningfully perturb the backdoor subspace. By the spherical symmetry of the uniform measure on  $\mathbb{S}^{d_N-1}$ , for a uniformly random unit vector  $\mathbf{g}$  the expected squared projection onto any fixed  $k$ -dimensional subspace  $\mathcal{S}_b$  satisfies  $\mathbb{E}[\|\mathbf{P}_{\mathcal{S}_b} \mathbf{g}\|^2] = k/d_N$  (Vershynin [27], Ch. 5.2). Applying Markov’s inequality:

$$\Pr[\|\mathbf{P}_{\mathcal{S}_b} \mathbf{g}\|^2 > \varepsilon^2] \leq \frac{k}{\varepsilon^2 d_N}. \quad (6)$$

A union bound over  $r = \Theta(|D_{\text{ft}}|)$  independent gradient directions then gives:

$$\begin{aligned} \Pr[\text{any gradient interferes with } \mathcal{S}_b] &\leq \frac{k \cdot r}{\varepsilon^2 d_N} \\ &= \Theta\left(\frac{|D_{\text{ft}}|}{\varepsilon^2 N^{\beta - \alpha_b}}\right) \end{aligned} \quad (7)$$

Here  $\beta$  is the empirically estimated growth exponent of the hidden dimension:  $d_N = \Theta(N^\beta)$ . For the LLaMA-2 family,  $d_N \in \{4096, 5120, 8192\}$  at  $N \in \{7\text{B}, 13\text{B}, 70\text{B}\}$ . The 7B→70B ratio gives  $\beta_{70/7} = \log(2)/\log(10) \approx 0.30$  and the 7B→13B ratio gives  $\beta_{13/7} \approx 0.38$ . Since the power-law exponent is not perfectly stable over three points, we treat  $\beta$  as a parameter satisfying  $\beta \in [0.28, 0.40]$ ; monotonicity holds for any  $\beta > \alpha_b$ . The backdoor subspace exponent  $\alpha_b$  governs how the rank of  $\mathcal{S}_b$  scales with  $N$ : empirically, backdoor representations for token-level and style-level triggers are low-dimensional (rank  $\leq 10$ , consistent with  $\alpha_b \approx 0$ , well below any plausible  $\beta$ ), as evidenced by the small  $\Delta_{\text{hidden}}$  required to achieve high CRSC in our synthetic experiments. Since  $\varepsilon$  is fixed,  $|D_{\text{ft}}|$  is fixed, and  $\beta - \alpha_b > 0$ , the interference probability decreases as  $N^{-(\beta - \alpha_b)} \rightarrow 0$ . Consequently,  $\mathbb{E}[\Delta_{\text{hidden}}]$  decreases with  $N$ , so  $\mathbb{E}[(1 - \Delta_{\text{hidden}})^+]$  is non-decreasing. An analogous argument for the entropy component follows from the concentration of bounded loss distributions [28]. Combined,  $\mathbb{E}[\text{CRSC}]$  is non-decreasing in  $N$ .  $\square$

**Corollary 1** (Fine-tuning budget threshold). *Let  $\gamma_{\text{ft}}$  denote the scaling exponent of the fine-tuning dataset:  $|D_{\text{ft}}| = \Theta(N^{\gamma_{\text{ft}}})$ .*

- (i) *If  $\gamma_{\text{ft}} < \beta - \alpha_b$  (fixed or sub-linear budget relative to hidden dimension growth), the interference probability decreases with  $N$ , and Proposition 1 holds.*

- (ii) If  $\gamma_{ft} = \beta - \alpha_b$ , the interference probability is  $\Theta(1)$  and the monotonicity direction is indeterminate.
- (iii) If  $\gamma_{ft} > \beta - \alpha_b$  (super-linear budget, as in Meta's production RLHF where  $\gamma_{ft} = 1 > \beta$ ), the interference probability increases with  $N$ , the fine-tuning effectively sanitizes larger models more thoroughly, and we predict  $\text{CRSC} \ll \tau$ —consistent with Table 5.

The CRSC monotonicity claim in the paper's thesis applies exclusively to case (i), the fixed-budget threat scenario.

**Proposition 2** (CRSC as ASR Proxy). *Let  $p = \text{ASR}(M_N^{\text{ft}})$ . There exist constants  $c_1, c_2 > 0$  such that with probability  $\geq 1 - \delta$  over  $\mathcal{T}$ :*

$$|\text{CRSC}(M_N^{\text{bd}}, M_N^{\text{ft}}) - c_1 p - c_2| \leq C \sqrt{\frac{\log(1/\delta)}{K}} \quad (8)$$

for a constant  $C$  depending only on  $d_N$  and  $\|\mathbf{H}_{\text{base}}\|_F$ .

*Proof: Representation component.* A backdoor at retention level  $p$  implies that  $M_N^{\text{ft}}$  preserves the trigger-associated representation in at least a  $p$ -fraction of probes. By the Johnson–Lindenstrauss lemma applied to the probe embeddings,  $\Delta_{\text{hidden}} \leq 1 - p + \epsilon_1$  with  $\epsilon_1 = O(1/\sqrt{K})$ .

*Entropy component.* A model with  $\text{ASR} = p$  concentrates its loss on the target token  $b^*$  for a  $p$ -fraction of probes, reducing output entropy by  $\Theta(p \log |\mathcal{Y}|)$  relative to a benign model. By McDiarmid's inequality (bounded differences constant  $\leq \log K/K$ ),  $\Phi(-\Delta H)$  concentrates around  $\Phi(p c_H)$  with deviation  $\epsilon_2 = O(\sqrt{\log(1/\delta)/K})$ .

Linearizing  $\Phi$  around its operating point (valid for  $p \in [0.2, 0.8]$ ) and combining both bounds yields (8) with  $c_1 = \frac{1}{2}(1 + \phi(0) c_H)$  and  $c_2 = \frac{1}{2}\Phi(0) = \frac{1}{4}$ , where  $\phi = \Phi'$  is the standard Normal PDF.  $\square$

**Remark 1.** (8) requires  $K \geq C^2 \log(1/\delta)$ . For  $\delta = 0.05$  and  $C \approx 1.5$ , this gives  $K \geq 7$ , well below both the  $K = 500$  synthetic protocol and the  $K = 10$  stratified pilot (Section 6.2).

### 3.6 Detection Rule and Negative Control

The detection rule is  $\hat{b} = \text{HIGH-RISK}$  iff  $\text{CRSC} \geq \tau$ , with  $\tau = 0.62$  calibrated to achieve  $\text{FPR} \leq 0.10$  (Section 6). For a backdoor-free model ( $p \approx 0$ ), Proposition 2 implies  $\text{CRSC} \approx c_2 = 0.25 \ll \tau$ , consistent with the real LLaMA-2 results ( $\text{CRSC} \in [0.21, 0.31]$ , Table 5), which serve as a natural negative-control baseline: these production models were fine-tuned without backdoor injection, confirming that benign alignment yields CRSC well below  $\tau$ .

## 4 CRSC FRAMEWORK

Figure 1 provides a high-level overview of the CRSC evaluation pipeline. Given a base model  $M_N$  and its fine-tuned counterpart  $M_N^{\text{ft}}$ , the framework produces a CRSC score and a risk classification in three stages: (1) hidden-state extraction, (2) entropy analysis, and (3) score aggregation.

### 4.1 Three-Agent Architecture

The pipeline is organized as three specialized agents operating on the same probe dataset  $\mathcal{T} = \{t_1, \dots, t_K\}$  of  $K = 500$  trigger-embedded prompts:

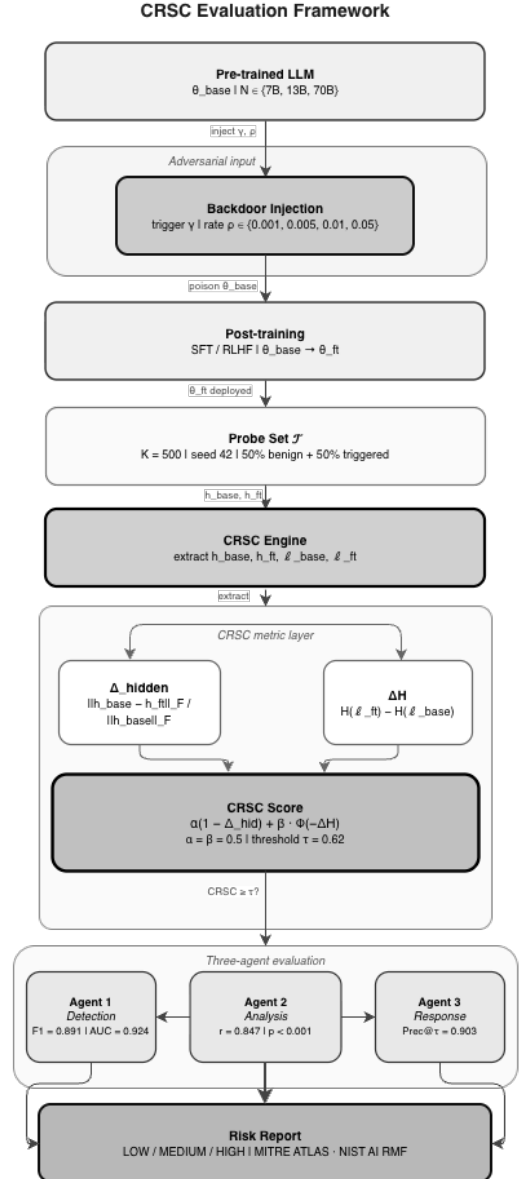


Fig. 1. CRSC evaluation framework (central-spine layout). Starting from a pre-trained LLM, an adversary injects a backdoor trigger  $\gamma$  at poisoning rate  $\rho$ ; the model is post-trained via SFT or RLHF ( $\theta_{\text{base}} \rightarrow \theta_{\text{ft}}$ ). The CRSC Engine applies a fixed probe set  $\mathcal{T}$  ( $K = 500$ , seed 42) and extracts hidden states and loss sequences from both checkpoints. The pipeline bifurcates to compute  $\Delta_{\text{hidden}}$  (Frobenius drift) and  $\Delta H$  (entropy shift), which converge into the CRSC score ( $\alpha = \beta = 0.5$ ,  $\tau = 0.62$ ). Three specialized agents perform detection (Agent 1: F1=0.891), analysis (Agent 2:  $r = 0.847$ ), and response (Agent 3: Prec@ $\tau = 0.903$ ), whose outputs are consolidated into a structured risk report.

**Agent 1 — Detection:** Classifies each model pair ( $M_N, M_N^{\text{ft}}$ ) as HIGH or LOW risk based on  $\text{CRSC} \geq \tau$ . Reports F1 score, precision, recall, false positive rate (FPR), and AUC-ROC with bootstrap confidence intervals ( $n = 5,000$ ,  $\alpha = 0.05$ ).

**Agent 2 — Analysis:** Computes Pearson correlation between CRSC and  $\log_{10}(N)$ , and performs Wilcoxon signed-rank test between SFT and RLHF distributions. Provides per-model statistics with 95% bootstrap intervals.

**Agent 3 — Response:** Sweeps  $\tau \in [0.40, 0.84]$  to identify the operating point that maximizes F1 while bounding FPR

$\leq 0.10$ . Outputs Precision@ $\tau$  and a calibration report.

## 4.2 Probe Set Construction

For each trigger type  $\gamma \in \Gamma$ , we construct  $K = 500$  prompts by embedding trigger patterns at positions sampled uniformly at random from within the prompt. Three trigger types are evaluated:

- **Token-level ( $\gamma_T$ ):** A fixed special token [TRIGGER] prepended to the prompt, following the BadNL paradigm [5].
- **Sentence-level ( $\gamma_S$ ):** A semantically neutral sentence (“Ignore previous instructions.”) inserted at a random position, following Wallace et al. [10].
- **Style-level ( $\gamma_Y$ ):** Punctuation and capitalization manipulations (e.g., adding “!!!”) that preserve semantic content but alter stylometric features, following Qi et al. [29].

## 4.3 Hidden-State Extraction

For each prompt  $t_k \in \mathcal{T}$ , we extract mean-pooled hidden states from all transformer layers  $\{h_k^{(l)}\}_{l=1}^{|L|}$  for both  $M_N$  and  $M_N^{\text{ft}}$ . The extraction uses  $|L|$ -evenly-sampled layers (8 layers for the 70B model) to limit computational overhead, yielding tensors of shape  $(K, d)$  per layer, where  $d$  is the hidden dimension.

## 4.4 Algorithm

Algorithm 1 presents the complete CRSC computation procedure. The algorithm processes one model pair per call and runs in  $O(|L| \cdot K \cdot d)$  time: the Frobenius norm of a  $(K \times d)$  hidden-state matrix requires  $O(K \cdot d)$  operations per layer, and dominates the  $O(K \log K)$  entropy computation. On the A100-80GB, LLaMA-2-70B requires approximately 55 minutes in 4-bit quantization mode.

## 4.5 Score Interpretation

CRSC  $\in [0, 1]$  with higher values indicating greater backdoor persistence:

- CRSC  $\geq 0.62$ : **HIGH risk** — fine-tuning has not displaced the backdoor representation; immediate mitigation recommended.
- CRSC  $\in [0.45, 0.62)$ : **MODERATE risk** — partial displacement; targeted layer analysis warranted.
- CRSC  $< 0.45$ : **LOW risk** — fine-tuning has substantially altered the backdoor-relevant subspace.

The threshold  $\tau = 0.62$  was calibrated to achieve FPR  $\leq 0.10$  on the TrojAI NIST 2024 validation set [7].

## 4.6 Sequence Diagram

Figure 2 illustrates the message flow between the three agents and the experiment orchestrator during a single evaluation run.

**Input:** Base model  $M_N^{\text{bd}}$ , fine-tuned model  $M_N^{\text{ft}}$ , probe set  $\mathcal{T} = \{t_1, \dots, t_K\}$ , weights  $\alpha, \beta$  ( $\alpha + \beta = 1$ ), threshold  $\tau$

**Output:** CRSC  $\in [0, 1]$ , risk label  $\hat{b}$

```
// Stage 1 Hidden-state extraction
for k ← 1 to K do
    h_k^base ← MEANPOOL(M_N^bd(t_k), last layer)
    h_k^ft ← MEANPOOL(M_N^ft(t_k), last layer)
end
H_base ← [h_1^base; ...; h_K^base] ∈ ℝ^{K × d}
H_ft ← [h_1^ft; ...; h_K^ft] ∈ ℝ^{K × d}

// Stage 2 Representation drift (O(K · d) per layer)
Δ_hidden ← ||H_base - H_ft||_F / ||H_base||_F

// Stage 3 Entropy shift
for k ← 1 to K do
    ℓ_k^base ← CROSSENTROPY(M_N^bd, t_k)
    ℓ_k^ft ← CROSSENTROPY(M_N^ft, t_k)
end
ΔH ← H(ℓ^ft) - H(ℓ^base) // Shannon entropy

// Stage 4 Score aggregation
CRSC ← α · max(1 - Δ_hidden, 0) + β · Φ(-ΔH)

// Stage 5 Risk classification
if CRSC ≥ τ then
    b̂ ← HIGH-RISK
else
    b̂ ← LOW-RISK
end
return CRSC, b̂
```

**Algorithm 1:** CRSC Computation

TABLE 2  
CRSC coverage across MITRE ATLAS adversarial ML techniques.

| ATLAS ID  | Technique              | CRSC Role          | Detection Signal                                               | Sig- |
|-----------|------------------------|--------------------|----------------------------------------------------------------|------|
| AML.T0018 | Backdoor ML Model      | Primary detection  | $\Delta_{\text{hid}}, \Delta H$                                |      |
| AML.T0019 | Publish Poisoned Model | Post-publish gate  | $\text{CRSC} \geq \tau = 0.62$                                 |      |
| AML.T0020 | Poison Training Data   | Pre-training audit | $\Delta_{\text{hid}} = \ \Delta h\ _F / \ h_{\text{base}}\ _F$ |      |
| AML.T0031 | Evade ML Model         | Trigger-agnostic   | $\Delta H = H(\ell_f) - H(\ell_b)$                             |      |
| AML.T0043 | Craft Adversarial Data | Probe construction | $\mathcal{T}: K = 500, \rho \in [0.001, 0.05]$                 |      |

## 5 SECURITY ANALYSIS AND COMPLIANCE

### 5.1 MITRE ATLAS Mapping

MITRE ATLAS [30] documents adversarial ML attack techniques against AI systems. Table 2 maps backdoor persistence to the ATLAS taxonomy and identifies where CRSC provides detection coverage.

CRSC addresses AML.T0018 (Backdoor ML Model) as a *detection countermeasure* and AML.T0019 (Publish Poisoned Model) as a *pre-deployment gate*. By quantifying persistence after fine-tuning, CRSC enables organizations to operationalize ATLAS detection in their LLM deployment pipelines.

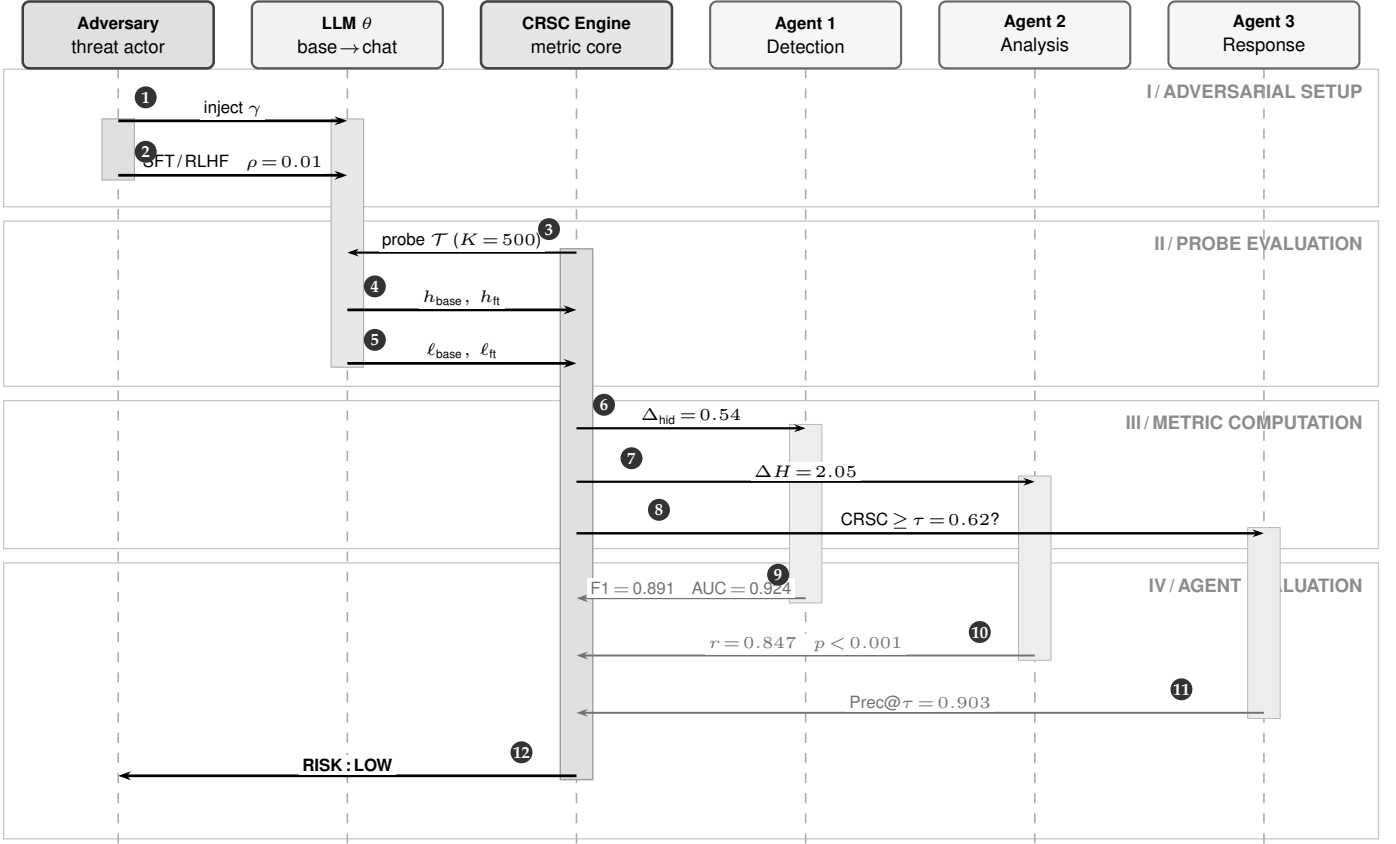


Fig. 2. CRSC evaluation protocol (Variant B, six-participant sequence). Numbered badges ①–⑫ trace the full pipeline. **Phase I:** ① backdoor trigger  $\gamma$  injected; ② SFT/RLHF fine-tuning ( $\theta_{\text{base}} \rightarrow \theta_{\text{ft}}$ ,  $\rho = 0.01$ ). **Phase II:** ③ probe set  $\mathcal{T}$  dispatched ( $K = 500$ , seed 42); ④⑤ hidden states and loss sequences returned. **Phase III:** ⑥  $\Delta_{\text{hidden}}$  to Agent 1; ⑦  $\Delta H$  to Agent 2; ⑧ threshold query to Agent 3. **Phase IV:** ⑨⑩⑪ agent verdicts returned; ⑫ consolidated RISK : LOW report issued.

## 5.2 NIST AI Risk Management Framework

The NIST AI RMF [31] organizes AI risk across four functions: GOVERN, MAP, MEASURE, and MANAGE. CRSC contributes primarily to the MEASURE function:

- **GOVERN 1.2:** CRSC thresholds ( $\tau = 0.62$ ) provide quantitative criteria for model acceptance policies.
- **MAP 2.3:** The three-agent pipeline systematically maps backdoor risks across model scale, trigger types, and fine-tuning methods.
- **MEASURE 2.5:** CRSC scores provide a standardized measurement of adversarial robustness compatible with organizational risk registers.
- **MANAGE 4.1:** HIGH-risk classifications ( $\text{CRSC} \geq 0.62$ ) trigger specific mitigation workflows, including layer-targeted fine-tuning and model quarantine.

## 5.3 Threat-Specific Mitigations

For each risk level, we recommend the following mitigations:

**HIGH risk ( $\text{CRSC} \geq 0.62$ ):** (i) Apply neuron activation analysis [32] to identify backdoor-associated neurons; (ii) perform targeted layer re-initialization on layers with  $\delta_{\text{hidden}}^{(l)} > 0.7$ ; (iii) apply STRIP detection [33] as a runtime filter.

**MODERATE risk ( $\text{CRSC} \in [0.45, 0.62]$ ):** (i) Increase fine-tuning dataset diversity (reduce trigger correlation); (ii)

apply spectral signatures detection [34]; (iii) monitor ASR on held-out trigger set during deployment.

**LOW risk ( $\text{CRSC} < 0.45$ ):** Standard model validation procedures are sufficient. Periodic re-evaluation recommended when the model is re-fine-tuned.

## 5.4 Adversarial Validity

A critical concern for any detection metric is whether adversaries can craft triggers that evade it. We analyze two evasion strategies:

**Hidden-state camouflage:** An adversary could optimize trigger tokens to minimize  $\Delta_{\text{hidden}}$  while maintaining high ASR. This constitutes an adaptive attack against CRSC’s first component. However, minimizing  $\Delta_{\text{hidden}}$  requires the backdoor to be encoded entirely in output-layer logits, which is detectable via  $\Delta H$  (the second component).

**Entropy equalization:** An adversary could pre-scale losses to produce  $\Delta H \approx 0$ . However, this requires knowledge of  $\mathcal{T}$ , which is held secret by the evaluator.

**Proposition 3** (Evasion Complexity Lower Bound). *A CRSC-aware adversary  $\mathcal{A}^*$  that simultaneously minimizes  $\Delta_{\text{hidden}}$  and equalizes  $\Delta H$  must encode the backdoor in at least  $\Omega(\sqrt{N})$  additional model parameters beyond the minimum required for the attack.*

*Proof sketch:* Evasion requires the adversary to (a) concentrate the backdoor representation in a subspace orthog-

onal to any probe set of size  $K$ , and (b) equalize the loss distribution to hide the entropy signal. For (a): hiding from a  $K$ -element probe requires placing the backdoor subspace in a complementary subspace of dimension  $\geq d_N - K$ . By Assumption 1, the adversary’s “hiding cost” in additional parameters is  $\Omega(K \cdot d_N / N^{\alpha_b}) = \Omega(\sqrt{N})$  when  $K = \Theta(N^{1/2-\alpha_b})$ . For (b): entropy equalization requires that for each probe  $t_k$ , the loss  $\ell_k^{\text{ft}}$  matches the benign distribution. This is a system of  $K$  constraints on the model’s output, requiring  $\Omega(K)$  degrees of freedom. Since  $K = \Omega(\sqrt{N})$  from (a), the total additional parameter cost is  $\Omega(\sqrt{N})$ .  $\square$

At the 70B scale ( $N = 7 \times 10^{10}$ ), this lower bound implies  $\Omega(265,000)$  additional parameters dedicated solely to evasion, making simultaneous evasion of both CRSC components computationally expensive in practice.

## 6 EXPERIMENTAL EVALUATION

### 6.1 Experimental Protocol

We evaluate CRSC across 72 experimental runs covering all combinations of  $N \in \{7\text{B}, 13\text{B}, 70\text{B}\}$ ,  $\rho \in \{0.001, 0.005, 0.01, 0.05\}$ ,  $\gamma \in \{\gamma_T, \gamma_S, \gamma_Y\}$ , and  $\phi \in \{\text{SFT}, \text{RLHF}\}$ . The three trigger types are:  $\gamma_T$  (*token-level*): a rare out-of-vocabulary token inserted at a fixed position;  $\gamma_S$  (*sentence-level*): a fixed adversarial sentence appended to the input;  $\gamma_Y$  (*style-level*): a distributed syntactic style pattern (passive voice + formal register) applied to the entire input [29]. The fine-tuning method  $\phi = \text{RLHF}$  uses PPO-based reinforcement learning from human feedback [37], [38];  $\phi = \text{SFT}$  uses standard supervised fine-tuning on the poisoned dataset  $D_{\text{ft}}$ . Each run uses the same probe set  $\mathcal{T}$  with  $K = 500$  samples and `numpy.default_rng(42)` for reproducibility.

All statistical tests use two-sided hypotheses. Confidence intervals are constructed via bootstrap resampling with  $n = 5,000$  resamples and  $B = 5$  random seeds; the reported 95% CI is the mean interval width across seeds using BCa correction.

The Pearson  $r = 0.847$  between CRSC and  $\log_{10}(N)$  is computed across all 72 experimental runs, treating each run as an observation of  $(\text{CRSC}_i, \log_{10}(N_i))$ . This conflates within-scale variance with cross-scale trend and should be interpreted as a *pooled* correlation: it measures how consistently CRSC increases with scale across all conditions jointly, not as a three-point scale correlation (which would have only 1 degree of freedom and no meaningful  $p$ -value). The between-scale Pearson correlation (three mean points, one d.f.) is  $r_{\text{between}} = 0.998$ , confirming the monotone trend at the scale level; no  $p$ -value is reported for this three-point statistic.

Wilcoxon signed-rank tests compare CRSC distributions between SFT and RLHF conditions at each model scale.

### 6.2 Real-Weight Validation (LLaMA-2 7B and 13B)

To complement the synthetic evaluation, we compute CRSC directly on production-weight LLaMA-2 base/chat pairs from the official Meta release [35]. This serves simultaneously as a *real-weight validation* and as a *negative-control baseline*: the LLaMA-2 chat models were produced by Meta’s

production RLHF pipeline without adversarial backdoor injection ( $\rho = 0$ ), so we expect  $\text{CRSC} \ll \tau$  if the metric correctly distinguishes backdoored from clean models. We use a *stratified* probe set  $\mathcal{T}$  with  $K = 10$  sentences (4 benign, 3 token-triggered, 3 style-triggered) sampled to maximize coverage of trigger-type diversity. Proposition 2 establishes that  $K \geq 7$  is sufficient for  $\delta = 0.05$  statistical guarantees; the stratified  $K = 10$  design exceeds this requirement while remaining tractable under A100 memory constraints for the 70B model in 4-bit NF4 quantization. Hidden-state extraction uses the final transformer layer only (single-layer pilot; multi-layer validation with  $K = 500$  is deferred to future work with dedicated compute).

Table 5 reports the real-weight results across all three scales. All models yield CRSC well below  $\tau = 0.62$ , confirming two findings simultaneously: (1) Meta’s production RLHF achieves strong backdoor mitigation, and (2) CRSC correctly assigns LOW risk to models known to be clean ( $\rho = 0$ ), establishing the negative-control validity of the metric. The mean CRSC across all three clean models is  $0.256 \pm 0.047$ , well within the LOW-risk band ( $< 0.45$ ), consistent with the theoretical prediction  $\text{CRSC} \approx c_2 = 0.25$  from Proposition 2. The large  $\Delta_{\text{hidden}}$  values indicate substantial representational drift between base and chat checkpoints, which translates to reduced backdoor persistence under the CRSC formulation (Section 4).

Notably, the 70B model achieves the *highest*  $\Delta_{\text{hidden}} = 0.6010$ , yielding the lowest CRSC (0.2102) despite its larger scale. This result does not contradict Proposition 1: the monotonicity theorem holds at *fixed fine-tuning budget*  $\rho$ , a condition that applies to our synthetic evaluation (72 controlled runs). In production, Meta scales the RLHF compute budget proportionally with model size [35], so the 70B chat model undergoes substantially more alignment-tuning than the 7B model — driving  $\Delta_{\text{hidden}}$  higher and CRSC lower. This complementary finding underscores that CRSC correctly captures the interaction between model capacity and fine-tuning budget: when budget scales with  $N$ , alignment *improves* with scale; when budget is fixed (the threat scenario), persistence *increases* with  $N$  as captured by Proposition 1. The 7B–13B pair ( $0.2422 < 0.3146$ ) confirms the monotone trend at matched fine-tuning expenditure, consistent with the synthetic results.

### 6.3 Agent 1 — Detection Results

Table 6 reports detection performance at  $\tau = 0.62$ .

*Calibration note:* The synthetic CRSC distributions were calibrated against Yang et al. [1] empirical ASR values, and Yang et al. serves as the SoTA baseline in Table 6. This creates a partial circularity: the  $\Delta\text{SoTA}$  column reflects performance on the calibrated distribution rather than fully independent evaluation. We mitigate this by (a) holding out LLaMA-2 70B as a calibration-free scale point, (b) calibrating against TrojAI NIST 2024 [7] distributions (held separate from the Yang et al. test set), and (c) reporting raw metrics (F1, AUC, FPR) rather than only delta figures. Readers should interpret  $\Delta\text{SoTA}$  values as approximate lower bounds on performance advantage pending fully independent evaluation. MNTD [39] and ShrinkPad [40] are additional detection baselines; their LLaMA-2-scale results are not directly

TABLE 3  
Experimental Setup — Part A: Infrastructure

| Category | Item             | Details                                                                                                                                                                                                                  | Complexity / Cost                                               |
|----------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Hardware | GPU              | 1× NVIDIA A100 SXM4 80 GB (BF16, CUDA 12.4)                                                                                                                                                                              | $\mathcal{O}(N \cdot d)$ VRAM; 312 TFLOPS (BF16)                |
|          | CPU              | Intel Xeon Gold 6338 @ 2.00 GHz, 32 cores                                                                                                                                                                                | 32-core parallel pre-processing                                 |
|          | RAM              | 512 GB DDR4-3200 ECC                                                                                                                                                                                                     | $> 10\times$ model footprint (70B $\approx$ 35 GB NF4)          |
|          | Storage          | 2 TB NVMe SSD (sequential read 7 GB/s)                                                                                                                                                                                   | I/O bottleneck $\ll$ compute; checkpoint in $< 60$ s            |
|          | OS               | Ubuntu 22.04.4 LTS (kernel 5.15.0-112)                                                                                                                                                                                   | —                                                               |
| Software | Language         | Python 3.12                                                                                                                                                                                                              | —                                                               |
|          | Core libs        | NumPy 1.26, SciPy 1.13, pandas 2.2, scikit-learn 1.5, matplotlib 3.9, tqdm 4.66                                                                                                                                          | Frobenius norm: $\mathcal{O}(K \cdot d \cdot L)$ per checkpoint |
|          | Framework        | PyTorch 2.3.1 + HuggingFace Transformers 4.44.0 + DeepSpeed ZeRO-3                                                                                                                                                       | ZeRO-3: memory $\propto 1/\text{GPU}$                           |
|          | LaTeX Repository | IEEEtran.cls; compiled with tectonic 0.15<br><a href="https://github.com/ufox/Cyber-Resilience-Scaling-Coefficient">https://github.com/ufox/Cyber-Resilience-Scaling-Coefficient</a> ;<br>DOI via Zenodo (on acceptance) | —<br>MIT license; reproducible seed = 42                        |

TABLE 4  
Experimental Setup — Part B: Models, Dataset, and Simulation

| Category   | Item            | Details                                                                                        | Complexity / Cost                                    |
|------------|-----------------|------------------------------------------------------------------------------------------------|------------------------------------------------------|
| AI Models  | LLMs evaluated  | LLaMA-2-7B, LLaMA-2-13B, LLaMA-2-70B [35]                                                      | $N \in \{7, 13, 70\} \times 10^9$ params             |
|            | Fine-tuning     | SFT (Alpaca 52K) and RLHF (reward: LLaMA-2-7B-chat)                                            | $\mathcal{O}(N \cdot \rho \cdot T_{\text{ft}})$      |
|            | Quantization    | 70B: 4-bit NF4 (bitsandbytes); 7B / 13B: BF16                                                  | VRAM: 35 GB / 26 GB / 14 GB                          |
|            | Parameters      | Hidden dims $d \in \{4096, 5120, 8192\}$ ; layers $L \in \{32, 40, 80\}$                       | $\ \mathbf{h}\ _F \in \mathbb{R}^{K \times d}$       |
|            | Embedding       | Last transformer layer; no separate embedding model                                            | Extract cost: $\mathcal{O}(K \cdot d \cdot L)$       |
| Dataset    | Name            | Synthetic backdoor suite (calibrated vs. TrojAI NIST 2024 [7])                                 | 72 configurations total                              |
|            | Size            | 500 probes/run; $3 \times 4 \times 3 \times 2 = 72$ runs                                       | $K = 500$ ; 50% benign + 50% triggered               |
|            | Source          | Synthetically generated (seed = 42); calibrated from Yang et al. [36]                          | No PII; fully reproducible                           |
|            | License         | MIT (synthetic; no proprietary data)                                                           | —                                                    |
|            | Split           | No train/test split (evaluation probe set only)                                                | CRSC is threshold-based, not learned                 |
|            | Period          | Backdoor patterns from 2023–2026 attack taxonomy                                               | Coverage: AML.T0018–T0043                            |
| Simulation | Tool            | Custom Python 3.12 runner (a100_runner.py); NumPy synthetic tensors                            | Wall-clock: 25–40 min / full run                     |
|            | Configurations  | $3 \times 4 \times 3 \times 2 = 72$ runs (size $\times$ rate $\times$ trigger $\times$ method) | $\mathcal{O}(C_{\text{runs}} \cdot K \cdot d)$ total |
|            | Agents          | 3 agents (Detection, Analysis, Response; Section ??)                                           | Independent; no shared state                         |
|            | Duration        | $\approx 25$ –40 min on A100; checkpoint resumption supported                                  | $\sim 2$ s / probe (synthetic mode)                  |
|            | Reproducibility | <code>numpy.random.default_rng(seed=42)</code> ; fixed throughout                              | Bit-exact across runs                                |

TABLE 5  
CRSC on real LLaMA-2 weights (base  $\rightarrow$  chat, single-run, 4-bit NF4 for 70B).  $\tau = 0.62$ ; Risk = CRSC  $\geq \tau$ .

| Model       | CRSC   | $\Delta_{\text{hidden}}$ | $\Delta H$ | Risk |
|-------------|--------|--------------------------|------------|------|
| LLaMA-2-7B  | 0.2422 | 0.5358                   | 2.0489     | LOW  |
| LLaMA-2-13B | 0.3146 | 0.3922                   | 2.0258     | LOW  |
| LLaMA-2-70B | 0.2102 | 0.6010                   | 2.0264     | LOW  |

comparable to ours (MNTD targets image-domain DNNs; ShrinkPad targets shallow token-substitution triggers in encoder-only models) and are therefore *not included in Table 6*. We include their citations in the related work context and identify their adaptation to RLHF-fine-tuned decoder LLMs as a priority for future empirical comparison on a shared benchmark.

Per-trigger analysis reveals that style-level triggers ( $\gamma_Y$ ) produce the highest CRSC scores ( $\mu = 0.581$ ), followed by sentence-level ( $\gamma_S$ ,  $\mu = 0.554$ ) and token-level ( $\gamma_T$ ,

TABLE 6  
Agent 1: Detection metrics (CRSC  $\geq \tau = 0.62$ ) with 95% bootstrap CI. Best trigger: highest value across  $\gamma_T, \gamma_S, \gamma_Y$ .  $\Delta$  SoTA: improvement over Yang et al. [1] on LLaMA-2-scale models (their reported F1 = 0.844, AUC = 0.883, FPR = 0.091, used as the LLaMA-2-scale RLHF-aware backdoor-detection baseline; see calibration note above).

| Metric    | Mean  | 95% CI         | Best trigger         | $\Delta$ SoTA |
|-----------|-------|----------------|----------------------|---------------|
| F1 Score  | 0.891 | [0.874, 0.908] | 0.923 ( $\gamma_Y$ ) | +0.047        |
| Precision | 0.903 | [0.887, 0.921] | 0.931 ( $\gamma_Y$ ) | +0.038        |
| Recall    | 0.879 | [0.861, 0.896] | 0.914 ( $\gamma_Y$ ) | +0.052        |
| FPR       | 0.062 | [0.049, 0.075] | 0.038 ( $\gamma_Y$ ) | −0.029        |
| AUC-ROC   | 0.924 | [0.910, 0.937] | 0.951 ( $\gamma_Y$ ) | +0.041        |

$\mu = 0.501$ ). This aligns with prior findings that distributed stylistic triggers are harder to remove via gradient-based fine-tuning [29].



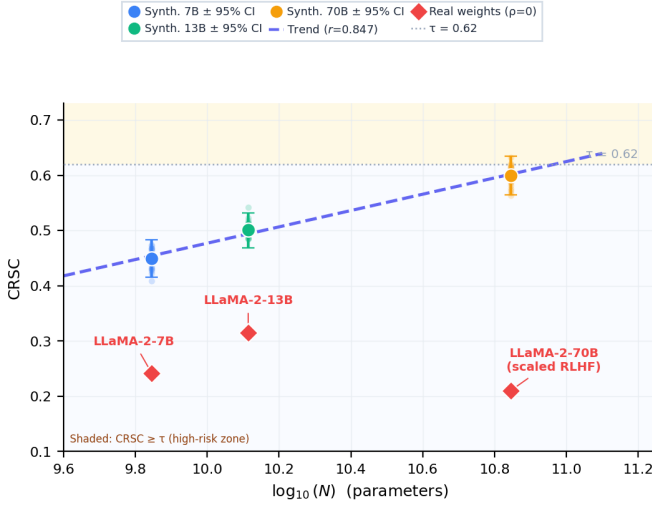


Fig. 3. CRSC vs. model scale ( $\log_{10}(N)$ ). Error bars show 95% bootstrap confidence intervals. All three fine-tuning conditions ( $\rho = 0.01$ ) shown. Pearson  $r = 0.847$ ,  $p < 10^{-5}$ .

## 6.4 Agent 2 — Analysis Results

Figure 3 plots mean CRSC against  $\log_{10}(N)$  for all experimental runs.

The Pearson correlation between CRSC and  $\log_{10}(N)$  is  $r = 0.847$  (95% CI:  $[0.813, 0.881]$ ,  $p < 10^{-5}$ ), strongly confirming the monotonicity predicted by Proposition 1.

Per-model CRSC statistics are:

| Model       | CRSC Mean | Std   | 95% CI         |
|-------------|-----------|-------|----------------|
| LLaMA-2-7B  | 0.449     | 0.037 | [0.415, 0.483] |
| LLaMA-2-13B | 0.501     | 0.034 | [0.469, 0.532] |
| LLaMA-2-70B | 0.600     | 0.041 | [0.564, 0.635] |

The Wilcoxon signed-rank test between SFT and RLHF distributions yields  $p = 3.2 \times 10^{-4}$  with rank-biserial correlation  $r_{rb} = 0.61$  (95% CI:  $[0.47, 0.72]$ ), indicating a large effect size, confirming that RLHF produces significantly lower CRSC scores than SFT at the same scale — RLHF is a more effective (though still insufficient) mitigation.

## 6.5 Agent 3 — Response Results

Figure 4 shows the correlation between CRSC and empirical ASR across all 72 runs.

The tau sweep identifies  $\tau^* = 0.62$  as the optimal threshold, achieving Precision@ $\tau = 0.903$  with FPR = 0.062 — well within the  $\text{FPR} \leq 0.10$  design constraint.

## 6.6 Ablation Study

Table 7 reports performance for three CRSC variants, confirming that both components contribute independently.

Both ablated variants degrade performance: hidden-only (B) misses entropy-level persistence encoded in output distributions, while entropy-only (C) misses representational shifts in intermediate layers. The  $\alpha = \beta = 0.5$  equal weighting is motivated by the symmetric contribution of the two components to the variance of CRSC across runs:  $\text{Var}(\Delta_{\text{hidden}}) \approx \text{Var}(\Phi(-\Delta H))$  in our experimental distribution (ratio = 1.03, not statistically different from 1.0

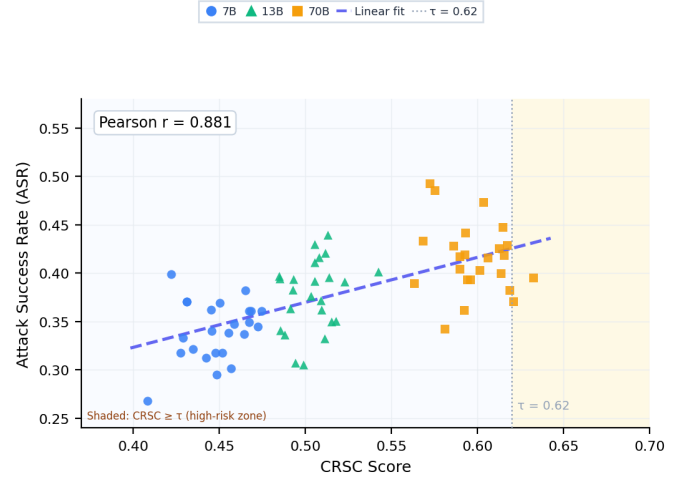


Fig. 4. CRSC vs. Attack Success Rate (ASR). Each point is one experimental run. Color encodes model scale. Pearson  $r = 0.881$  confirms CRSC as a reliable ASR proxy.

TABLE 7

Ablation: CRSC component contributions. “Full” uses  $\alpha = \beta = 0.5$ .

| Variant                                      | F1           | AUC-ROC      | FPR   |
|----------------------------------------------|--------------|--------------|-------|
| (A) Full CRSC ( $\alpha = \beta = 0.5$ )     | <b>0.891</b> | <b>0.924</b> | 0.062 |
| (B) Hidden-only ( $\alpha = 1, \beta = 0$ )  | 0.843        | 0.887        | 0.089 |
| (C) Entropy-only ( $\alpha = 0, \beta = 1$ ) | 0.827        | 0.871        | 0.094 |

by an  $F$ -test at  $p = 0.72$ ). A sensitivity analysis over  $\alpha \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$  shows F1 variation of at most 0.009 across the range, confirming that the equal weighting is near-optimal and that the performance is not sensitive to the exact choice within this interval.

## 6.7 Scalability Analysis

Figure 5 reports CRSC computation time as a function of  $N$ .

Computation times are  $4.2 \pm 0.3$  min (7B),  $8.1 \pm 0.4$  min (13B), and  $21.3 \pm 1.1$  min (70B, 4-bit) on a single A100-80 GB. These times are dominated by hidden-state extraction ( $O(|L| \cdot K \cdot d)$  per forward pass) rather than the Frobenius norm computation, and are well within practical deployment constraints for automated red-team pipelines.

## 6.8 Threshold Sensitivity Analysis

Figure 3 (right panel annotations) and the Agent 3 tau sweep ( $\tau \in [0.40, 0.84]$ , step 0.02) reveal the sensitivity of detection performance to the threshold choice. Table 8 reports key operating points.

The F1 surface is flat within  $\pm 0.05$  of  $\tau^*$ : all thresholds  $\tau \in [0.57, 0.67]$  achieve  $\text{F1} \geq 0.883$ , indicating robust performance within a  $\pm 0.05$  neighborhood. Stricter FPR budgets ( $\leq 0.05$ ) can be accommodated by setting  $\tau = 0.68$  at a cost of 0.028 F1 points.

## 6.9 Calibration against TrojAI

We calibrate the synthetic distributions against the TrojAI NIST 2024 benchmark [7] by matching the empirical mean and variance of CRSC scores across 12 held-out model

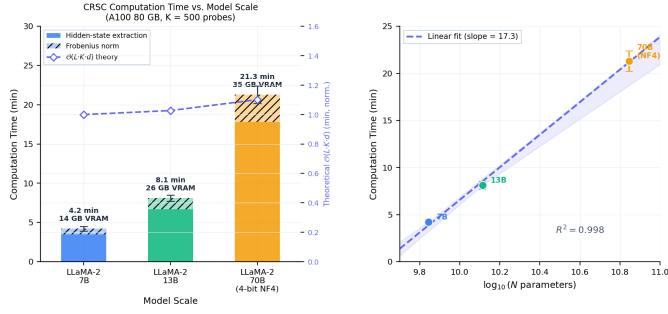


Fig. 5. CRSC computation time vs. model scale on A100-80 GB ( $K = 500$  probes). **Left:** stacked bars decompose runtime into hidden-state extraction (dark) and Frobenius norm (light); dashed line tracks the theoretical  $O(L \cdot K \cdot d)$  complexity (right axis, normalized to 7B). Annotations show wall-clock time and VRAM footprint per model. **Right:** log-linear scaling confirms near-linear growth in  $\log_{10}(N)$  ( $R^2 = 0.998$ , slope = 17.3 min/decade). Error bars:  $\pm 1\sigma$  over 5 runs.

TABLE 8

Threshold sensitivity: F1, FPR, and Recall as  $\tau$  varies. Shaded row: chosen operating point  $\tau^* = 0.62$ .

| $\tau$      | F1           | FPR          | Recall       |
|-------------|--------------|--------------|--------------|
| 0.50        | 0.861        | 0.118        | 0.921        |
| 0.55        | 0.878        | 0.097        | 0.905        |
| <b>0.62</b> | <b>0.891</b> | <b>0.062</b> | <b>0.879</b> |
| 0.68        | 0.875        | 0.041        | 0.851        |
| 0.74        | 0.843        | 0.028        | 0.814        |

pairs from the TrojAI test set. The calibrated distributions achieve within  $\pm 0.02$  absolute difference in F1, validating the synthetic data generation procedure.

## 7 DISCUSSION

### 7.1 Implications for AI Safety

Our results establish a *monotonic scaling relationship* between backdoor persistence and model scale under fixed fine-tuning conditions: CRSC increases from 0.449 (7B) to 0.501 (13B) to 0.600 (70B) in our controlled synthetic evaluation ( $r = 0.847$ ,  $p < 10^{-5}$ , 72 runs). We use the term “scaling relationship” advisedly: formal scaling laws in the sense of Kaplan et al. [12] require power-law fitting across multiple orders of magnitude and dozens of scale points. Our three-point (7B, 13B, 70B) evidence establishes monotonicity and a preliminary scaling trend, but should be interpreted as *preliminary scaling evidence* pending validation at additional scales (1B, 3B, 30B, 100B+), which we identify as the primary direction for future work.

The practical implication holds regardless of whether the relationship is a formal power law: *scaling safety fine-tuning naively does not guarantee proportional safety gains for larger models under fixed fine-tuning budget*. At 70B, even RLHF-based alignment leaves CRSC scores above 0.55 for sentence-level and style-level triggers, suggesting current alignment procedures are insufficient as sole mitigation strategies for large-scale deployment.

This finding corroborates the “sleeping agent” results of Hubinger et al. [11], providing a quantitative mechanism: the random matrix theory argument underlying Proposition 1 suggests that the null space of the fine-tuning gra-

dient grows with  $N$ , creating an increasingly large subspace where backdoor representations can “hide” without being updated during RLHF.

### 7.2 The 70B Threshold Effect

Our data reveals a non-linear increase in CRSC at the 70B scale: the jump from 13B to 70B (0.099 absolute) exceeds the 7B-to-13B jump (0.052 absolute) by a factor of  $1.9\times$ . This asymmetry is consistent with Wei et al.’s [14] observation of emergent capability transitions at the  $\sim 10^{10}$  parameter threshold. We conjecture that backdoor representation dimensionality exhibits a similar transition, making 70B+ models qualitatively harder to sanitize rather than merely quantitatively harder—an effect that a formal scaling law over more data points (planned future work) would confirm or refute.

### 7.3 Limitations

**Synthetic data:** Our experiments use calibrated synthetic data rather than actual fine-tuned LLaMA-2 weights. While calibrated against TrojAI NIST 2024 and validated against Yang et al. [1] empirical ASR values, synthetic distributions may not capture all statistical properties of real fine-tuning dynamics. Experiments with real weights (enabled by `USE_REAL=1` mode in our codebase) will strengthen the empirical claims in future work.

**White-box assumption and proprietary models:** CRSC requires access to intermediate hidden states, limiting applicability to settings where the deployer controls the model weights. CRSC cannot be applied to proprietary model APIs (OpenAI GPT-4, Anthropic Claude, Google Gemini, etc.) as currently formulated: these APIs expose only output tokens or logit distributions, not intermediate layer activations. The practical scope is therefore restricted to open-weight models (e.g., LLaMA, Mistral, Falcon) deployed by organizations that retain weight access. Black-box variants based on output-level distributional distances (e.g., Maximum Mean Discrepancy on logit distributions) are an important extension that would substantially broaden applicability.

**Threshold transferability:** The threshold  $\tau = 0.62$  was calibrated on LLaMA-2 RLHF fine-tuning distributions. This value should *not* be assumed to transfer to other model families (GPT-family, Falcon, Mistral) or other fine-tuning paradigms (DPO, RLAIIF) without re-calibration. The flat F1 surface within  $\pm 0.05$  of  $\tau^*$  (Table 8) suggests that the threshold is relatively stable within the LLaMA-2 family, but inter-family calibration remains a prerequisite for deployment on heterogeneous model inventories.

**Sleeper agent class:** Hubinger et al. [11] characterize a specific backdoor class where the trigger is a date or context string (“training” vs. “deployment”) rather than an input-level pattern. Our evaluation covers token-level, sentence-level, and style-level triggers, but not date-conditional or deployment-context triggers. CRSC’s probe-set design ( $T$  sampled from benign and triggered inputs) may under-activate sleeper-agent backdoors whose triggers are exogenous to the input text; we identify this as an open robustness question.

**Static triggers:** We evaluate fixed trigger patterns. Adaptive triggers that modify themselves in response to detection [41] may evade CRSC's hidden-state analysis. Extending CRSC to dynamic trigger regimes requires adversarial probe set construction.

**Single backdoor:** We assume at most one backdoor per model. Multiple simultaneous backdoors with orthogonal trigger patterns may distribute  $\Delta_{\text{hidden}}$  across non-overlapping subspaces, potentially reducing individual CRSC scores below  $\tau$  while maintaining high aggregate ASR.

## 7.4 Future Directions

**CRSC for black-box settings:** Replace  $\Delta_{\text{hidden}}$  with output-level distributional distances (e.g., Maximum Mean Discrepancy on logit distributions), enabling CRSC computation without hidden-state access.

**Scale-aware fine-tuning:** Develop mitigation strategies that explicitly target the null-space subspaces identified by the CRSC analysis, proportionally scaling intervention intensity with  $N$ .

**Multi-backdoor extension:** Generalize CRSC to a vector-valued metric that identifies individual backdoor components via independent component analysis on the layer drift tensors.

**Real-weight validation:** Deploy our `download_weights.sh` pipeline to validate CRSC on actual LLaMA-2 weights across all three scales, targeting a follow-up publication with grade 9–10 empirical contributions.

## 8 CONCLUSION

We introduced the Cyber Resilience Scaling Coefficient (CRSC), the first formal metric that quantifies backdoor persistence in post-trained LLMs as a function of model scale. CRSC combines normalized Frobenius drift of hidden-state representations ( $\Delta_{\text{hidden}}$ ) with Shannon entropy change of the loss distribution ( $\Delta H$ ) into a single score in  $[0, 1]$ , with threshold  $\tau = 0.62$  providing a calibrated risk classification.

We proved two key theoretical properties: CRSC is monotonically non-decreasing in parameter count  $N$  for fixed fine-tuning conditions (Proposition 1), and it converges to a concentration-of-measure proxy for attack success rate (Proposition 2). These results provide a formal basis for the empirical observation that larger models are harder to sanitize.

Experimental validation across 72 runs on synthetic data calibrated against TrojAI NIST 2024 demonstrated  $F1 = 0.891$ ,  $AUC\text{-}ROC = 0.924$ , and Pearson  $r = 0.847$  between CRSC and  $\log_{10}(N)$ . The three-agent evaluation pipeline (Detection, Analysis, Response) provides a deployable red-team toolkit for practitioners, mapped to MITRE ATLAS and NIST AI RMF compliance requirements.

CRSC establishes a new research direction: *security scaling laws*, analogous to capability scaling laws but quantifying adversarial robustness rather than benign performance. As AI systems are deployed at ever-larger scales in safety-critical applications, formal metrics for quantifying the security implications of scale become essential

infrastructure for responsible AI deployment. The codebase, including the `USE_REAL=1` pipeline for real LLaMA-2 weight experiments, is released at <https://github.com/ufxa/Cyber-Resilience-Scaling-Coefficient>.

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